Toward Responsible Artificial Intelligence

CS3263
Foundations of Artificial Intelligence
Version 3.0

Topics

- Human Factors in AI
  - Toward Human-AI Collaboration
  - Cognitive Models and Judgmental Biases
- Responsible AI Principles
  - Transparency
  - Privacy
  - Fairness
  - Safety and Robustness
  - Governance and Regulation

Agentic Artificial Intelligence

Predictive AI

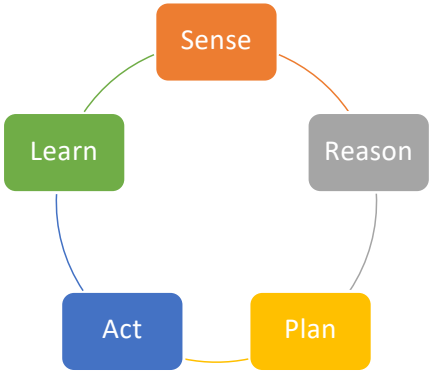
Classifies data. Makes forecasts.

Generative AI

Creates text, images, or plans from prompts.

Agentic AI

Plans, learns, uses tools, and acts to perform tasks.

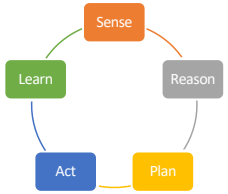


- Agentic AI (usually built on LFMs):
  - **Sense → Reason → Plan → Act → Learn**
  - Use tools, perform tasks, and adapt within guardrails
- Why It Matters:
  - Enables autonomous, generative decision-making
  - Fast, personalized, and coordinated across systems

Generative AI & Foundation Models

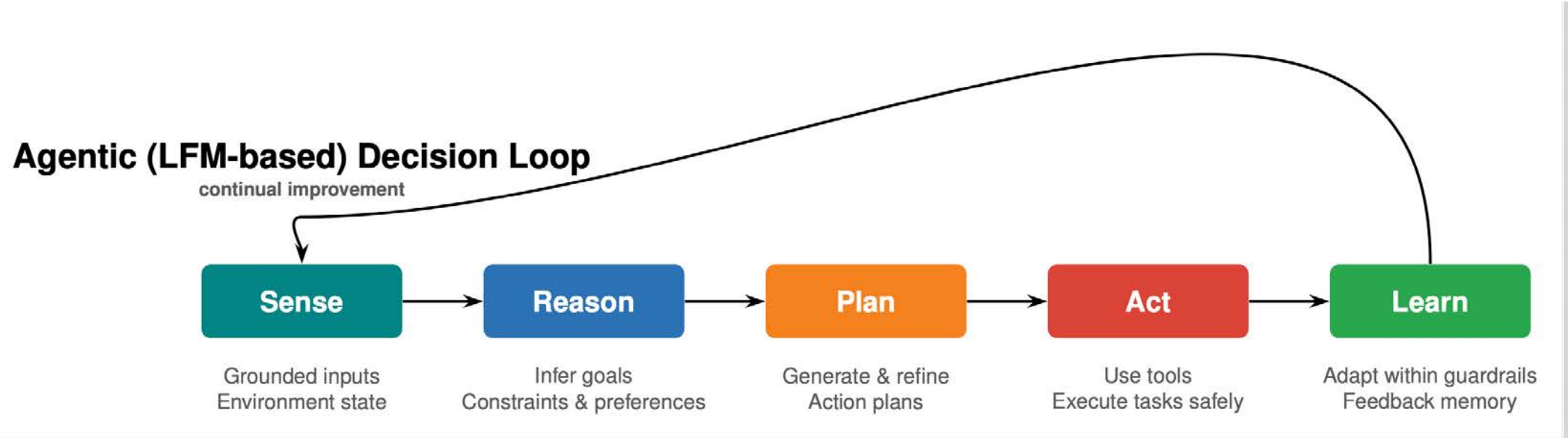
- What It Is
  - AI that generates original content — text, images, plans — by learning from data.
- Large Foundation Models (LFMs)
  - A class of generative AI models trained on vast data to generalize across tasks.
  - Examples: GPT, PaLM, LLaMA, DeepSeek, QWEN
- Key Capabilities
  - **Few-shot learning**: perform tasks with minimal examples
  - **Multimodal**: work with text, images, and other inputs
  - **Reasoning & planning**: support complex decisions in dynamic settings
  - **Scalable:** adapt across domains, from writing to strategic analysis
- Why It Matters
  - LFMs power the shift from generative outputs to agentic decision systems — **blending creation with action**.

A REAL Example



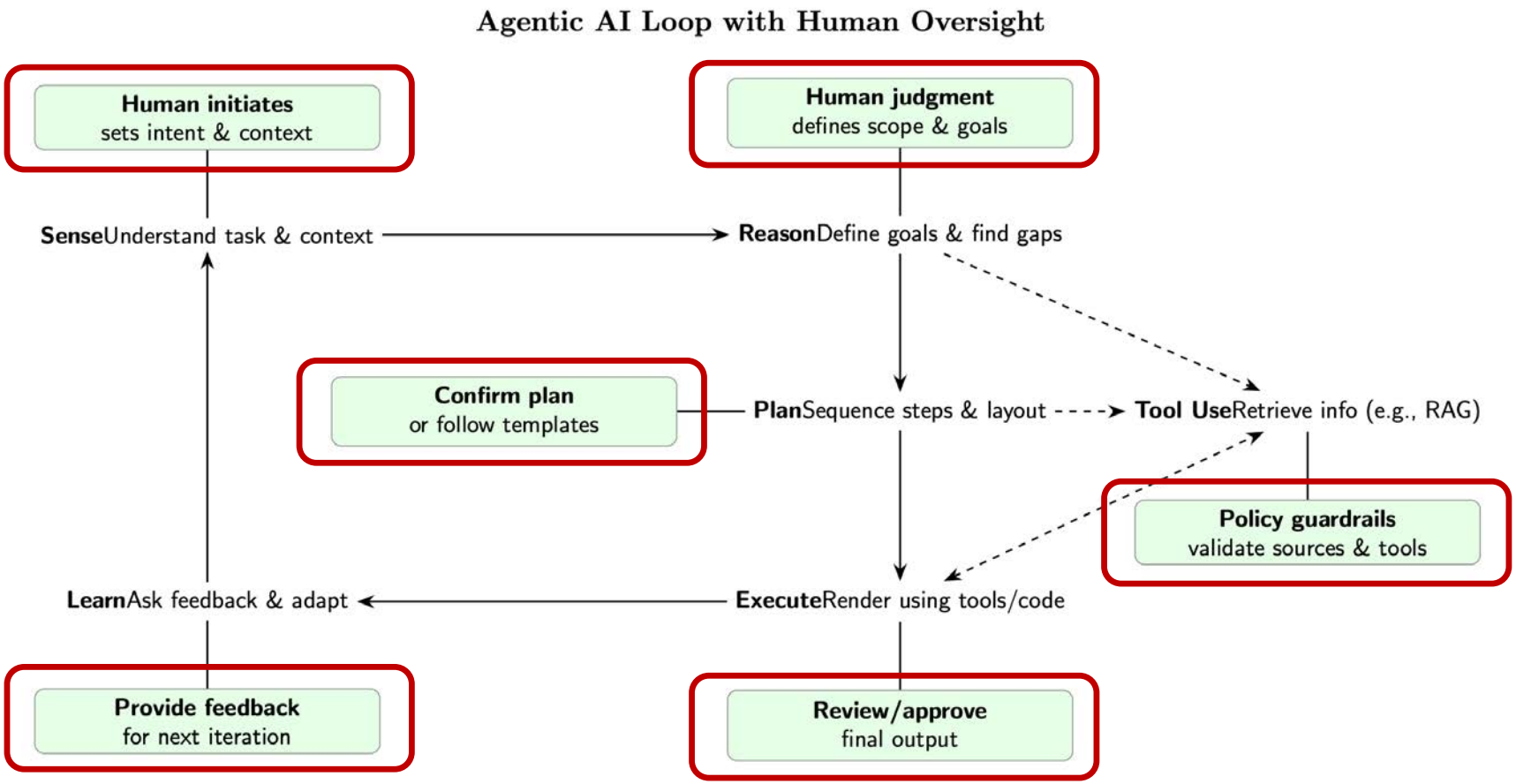
Agent(Learn)

LFM Generated Diagram from Latex



Source: ChatGPT 5.0 accessed 29 Oct 2025

Agentic Decision Loop with Human Oversight



Future of Artificial Intelligence

https://www.nature.com/articles/d41586-021-01170-0

https://www.technologyreview.com/2023/09/15/1079624/deepmind-inflection-generative-ai-whats-next-mustafa-suleyman/

Human Factors

Rethinking Rational Decision Making

- Rational decision making based on Maximum Expected Utility (MEU) principle
  - A rational agent should choose the action that maximizes its expected utility
- Are we rational?
  - Why is rational decision making difficult?
  - Will rational decisions and approaches always lead to good outcomes?
  - Are we usually rational?
- Humans are "predictively irrational!"
- Descriptive decision theory and cognitive modeling
  - Understanding judgmental or behavioral decision making
  - Relevant to ethical, governance and regulatory considerations in AI

Understanding Human Cognition

- Dual Process Theory
  - Human cognition operates through two distinct systems of thinking
  - These systems influence how we perceive, process information, and make decisions
  - Recognizing these systems helps us understand cognitive biases and improve decision-making
- System 1 Thinking: (Heuristic Processes)
  - Fast, automatic, effortless, implicit, and emotional
  - Provides speedy, efficient, and often quite accurate responses
- System 2 Thinking: (Analytic Processes)
  - Slower, conscious, effortful, explicit, and logical
  - Requires deliberate reasoning, time, and attention

Everyday decision making

Complex problem-solving and critical thinking

Simon, H. A. (1957). Models of man; social and rational. Oxford, England, Wiley
Stanovich, K. E., & West, R. F. (2000). Individual differences in reasoning: Implications for the rationality debate. Behavioral and Brain Sciences, 23(5), 645-665.
Kahneman, D. (2011). Thinking, Fast and Slow. New York: Farrar, Straus and Giroux.

Human Cognition and Judgmental Heuristics

- What are heuristics?
  - Simplifying strategies or rules of thumb
  - Simple ways of dealing with complex world
  - Usually provide acceptable results
- Common judgmental heuristics
  - Availability
  - Representativeness
  - Anchoring and adjustments
- Pitfalls:
  - Adopt heuristics without being aware of them
  - Misapplication to inappropriate situations leading to wrong/bad solutions

Types of judgment:
Prediction
Evaluation
Determination
Likelihood

Human Judgment and Bias in Agentic AI Alignment

Human–AI Command Loop

- Human feedback steers agentic AI reasoning
- Goal: Align AI behavior with human values
- Risk: Bias and inconsistency distort alignment

Bias & Judgment Risks

- **Cognitive Biases**: Anchoring, confirmation, availability
- **Evaluator Variability**: Inconsistent judgments across users
- **Over-Reliance** (Automation Bias): Blind trust in AI outputs
- **Under-Reliance** (Algorithm Aversion): Distrust of AI recommendations
- **Opaque Reliability**: Hard to gauge AI decision quality

Challenges and Opportunities

- Key Questions:
  - Which values should AI prioritize?
  - How do we handle conflicting values?
  - How do we adapt to changing data and preferences?
- Next Steps:
  - Align AI with public goals through transparency, regular updates, and stakeholder input.
  - Build Responsible AI systems that are human-aware and trustworthy!

Responsible Artificial Intelligence

Responsible Artificial Intelligence

- Toward Human-Aware AI Systems
  - AI working for, working with, and working alongside Humans
  - Supporting Human-AI collaboration
- Toward Trustworthy AI Systems
  - Natural interaction and effective communication
  - Fairness, accountability, and transparency
  - Robustness, resilience, privacy and security
  - Social, ethical, governance and regulatory considerations

Notes: Practical Applications

- Education: Ensure fair access to quality learning for all students.
- Social Services: Prioritize equal access for underserved groups.
- Healthcare: Align with societal values for patient safety and fairness.
- Law and Regulation: Ensure ethical, transparent AI in legal decision-making.
- Energy: Optimize resource distribution while supporting sustainability.
- Cybersecurity: Protect data privacy while upholding public safety.
- Urban Design: Plan infrastructure that meets community needs and inclusivity.

Common Principles of Responsible AI

How to incorporate these considerations into "rational" decision making?

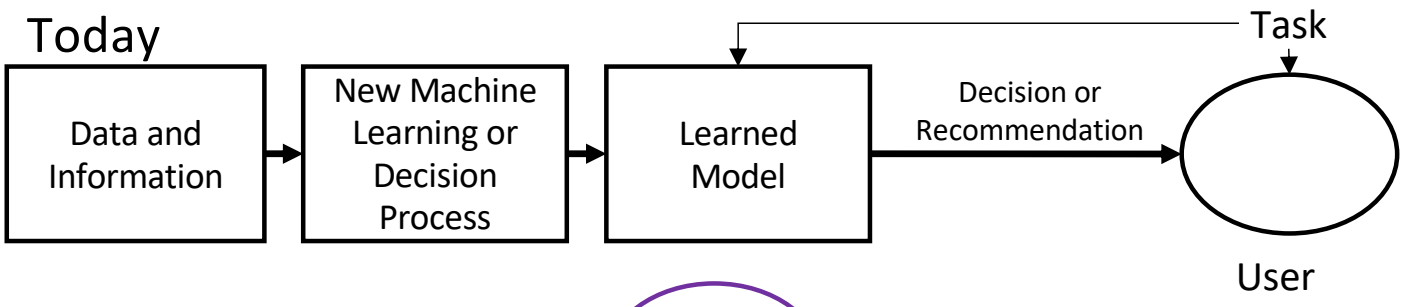
- Ensure safety
- Respect privacy
- Ensure fairness
- Promote trust
- Establish accountability
- Provide transparency
- Attribute responsibility
- Reflect diversity/inclusion
- Support equality
- Facilitate collaboration
- Uphold human rights and values
- Limit harmful uses of AI
- …

Ref: RN: Chapter 27

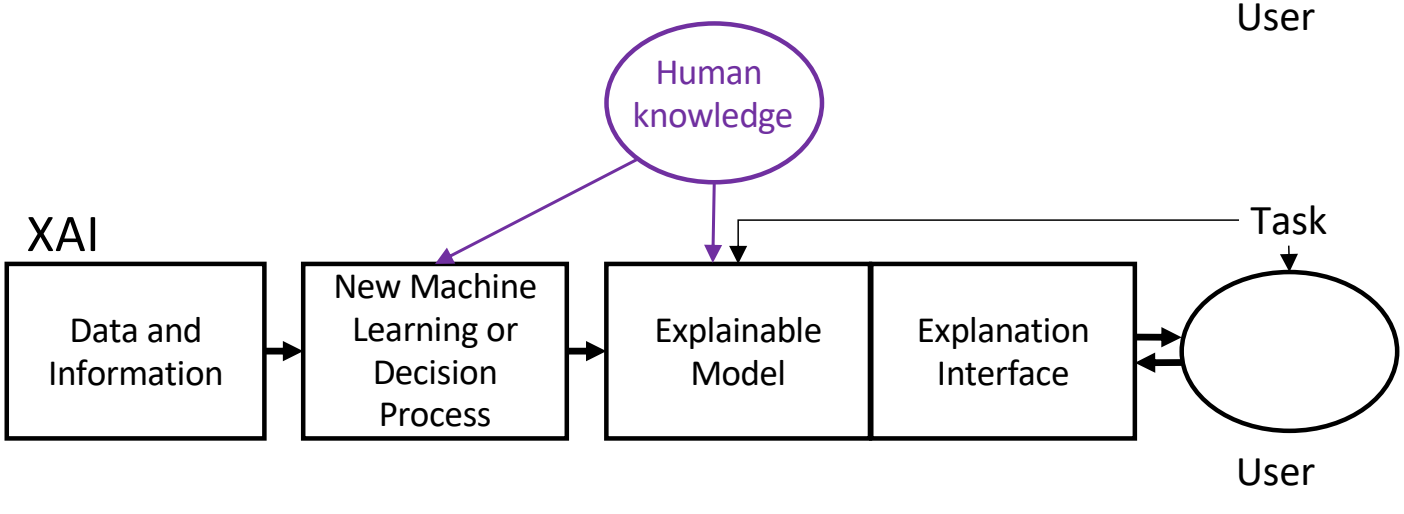
Transparency

Toward XAI – Explainable AI

Explainability
Vs
Interpretability



- Why did you do that?
- Why not something else?
- When do you succeed?
- When do you fail?
- When can I trust you?
- How do I correct an error?



- I understand why
- I understand why not
- I know when you succeed
- I know when you fail
- I know when to trust you
- I know why you erred

Adapted from: Resources on https://www.darpa.mil/program/explainable-artificial-intelligence

Toward Explainable AI …

Interpretable Models

Techniques to learn more structured, interpretable, causal models

Deep Explanation

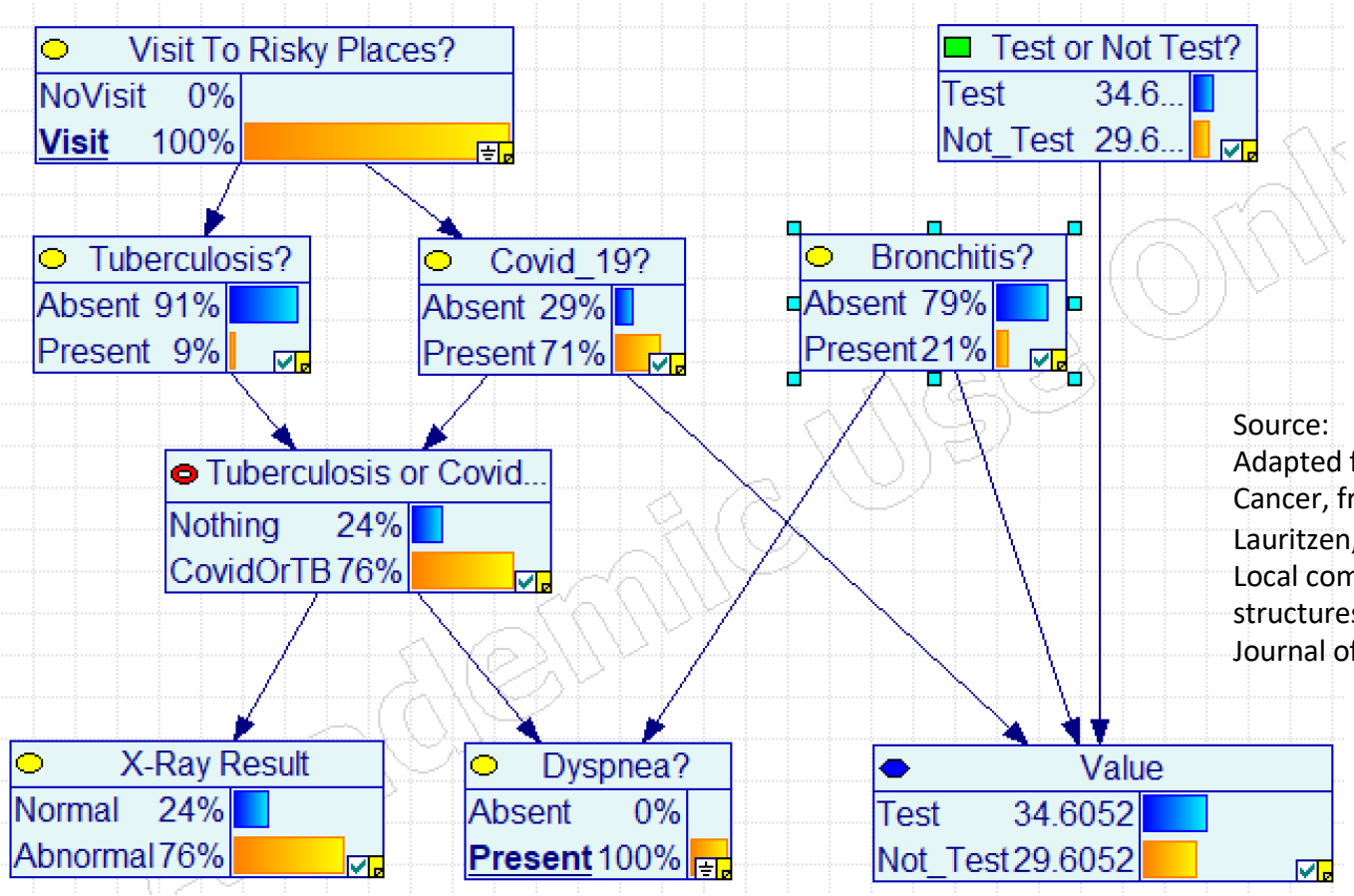
Modified deep learning techniques to learn explainable features

Model Induction

Techniques to infer an explainable model from any model as a black box

Adapted from: Resources on https://www.darpa.mil/program/explainable-artificial-intelligence

Notes: Graphical Probabilistic Models



Source:
Adapted from GENIE academic version Example – Lung Cancer, from the original network first appeared in:
Lauritzen, Steffen L. & Spiegelhalter, David J. (1988). Local computations with probabilities on graphical structures and their application to expert systems, Journal of the Royal Statistical Society B, 50(2):157-224

Transparency in AI Today

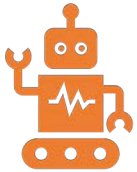
General Methodologies

Interpretable models
(preferred where feasible)

Post-hoc explanations
(with caution)

Faithfulness over persuasion

Audience-appropriate explanations



Generative AI Considerations

Training data opacity

Emergent behavior

System-level explanations

Human-in-the-loop oversight

Transparency Artifacts

Model Cards / System Cards

Datasheets for Datasets

Evaluation & robustness reports

Logging, traceability, auditability

Takeaway

Transparency is goes beyond explanations to support oversight and accountability across the AI lifecycle

Ref: RN Chapter 27; Lundberg and Lee, NIPS 2017, Refs: NIST AI RMF (2023); OECD AI Principles (2022); Doshi-Velez & Kim (2017)

Layers of Transparency

- Model transparency (architecture, limits)
- Data transparency (sources, bias)
- Decision transparency (local & global explanations)
- Process & accountability transparency

Privacy

Example: Privacy and Re-identification

Name	Gender	Age	Postal Code	Smoker	Diagnosis
Mei Mei	F	68	81317	N	Dementia
Susan	F	61	81382	N	Gastric Disease
Lee	F	67	81304	N	Arthritis
Seng	M	53	81359	Y	Heart Disease
James	M	59	81303	Y	Kidney Disease
Lily	F	52	81359	N	COVID-19
Tony	M	59	81330	Y	Diabetes, Hypertension
Cindy	F	55	81344	N	Lung Cancer

Source: Adapted from The Ethical Algorithm, Chapter 1, 2019

Example: Will K-Anonymity Help?

Suppressed Coarsened Coarsened

Name	Gender	Age	Postal Code	Smoker	Diagnosis
*	F	60-70	813**	N	Dementia
*	F	60-70	813**	N	Gastric Disease
*	M	60-70	813**	N	Arthritis
*	M	50-60	813**	Y	Heart Disease
*	M	50-60	813**	Y	Kidney Disease
*	F	50-60	813**	N	COVID-19
*	M	50-60	813**	Y	Diabetes, Hypertension
*	F	50-60	813**	N	Lung Cancer

Notes:

- k-anonymity: to redact information from individual records so that no set of characteristics matches just a single data record
- A released set of records is k-anonymous if any combination of insensitive attributes appearing in the database matches at least k individuals in the released data.

Source: Adapted from The Ethical Algorithm, Chapter 1, 2019

Problems with K-Anonymity and Others

- Re-identification is not the only privacy risk.
- Suppose we know that Lily is a 50+ patient at hospital A – either COVID-19 or lung cancer
- Still a serious privacy violation, cannot be prevented by k-anonymity
- Guarantees go away when multiple datasets are released – Lily went to both hospital A and B

Name	Gender	Age	Postal Code	Smoker	Diagnosis (A)
*	F	50-60	813**	N	COVID-19
*	F	50-60	813**	N	Lung Cancer

Name	Gender	Age	Postal Code	Smoker	Diagnosis (B)
*	F	50-60	813**	N	COVID-19
*	F	50-60	813**	N	Pancreatic Cancer
*	F	50-60	813**	N	Allergies

Source: Adapted from The Ethical Algorithm, Chapter 1, 2019

Privacy

Assumptions & Contexts

Secure systems and cybersecurity controls are in place

Privacy operates on top of data governance and access control frameworks

Regulatory contexts

HIPAA (USA, health data)

FERPA (USA, education data)

GDPR (EU)

PDPA (Singapore)

Traditional Privacy Techniques

De-identification and pseudonymization

Generalization of fields

k-anonymity and related notions

Aggregate querying and query limits

Limitation:
Protects against direct identification, not all privacy risks

Privacy in AI: Core Approaches

Differential privacy

Federated learning

Secure computation

Synthetic data

Limitation:
Privacy guarantees depend on assumptions and threat models.

Privacy in AI: Beyond Anonymization

Training data memorization

Membership inference attacks

Model inversion attacks

Secondary use and function creep

Key Takeaway

Privacy is a system-level property, not just a data preprocessing step.

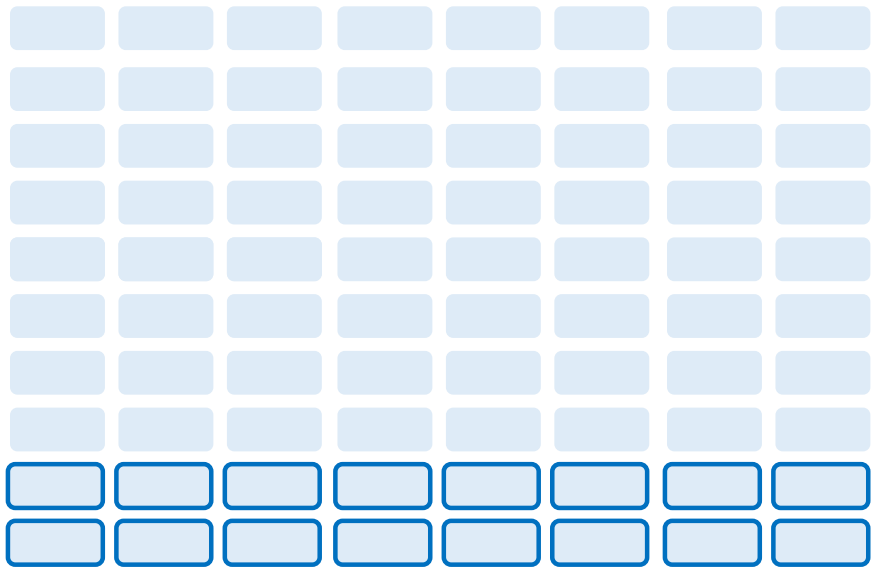
Refs: Russell & Norvig (2020); Dwork & Roth (2014); Shokri et al. (2017); Carlini et al. (2021); NIST AI RMF (2023)

Fairness

Example: Fairness in Context

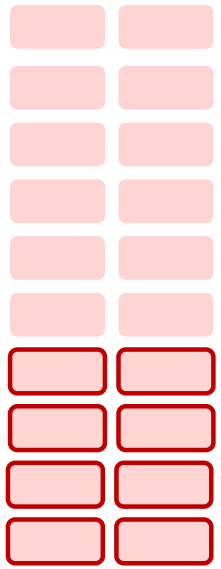
Protected Attribute:
Nose Color

- Consider 100 loan applications. AI approves or rejects loan applications fairly.
- 80 applicants have **blue nose** and 20 have **red nose**



20% (16)
Blue Nose

80 loan applicants,
20% will pay back



40% (8)
Red Nose

20 loan applicants,
40% will pay back

Example: Fairness in Context

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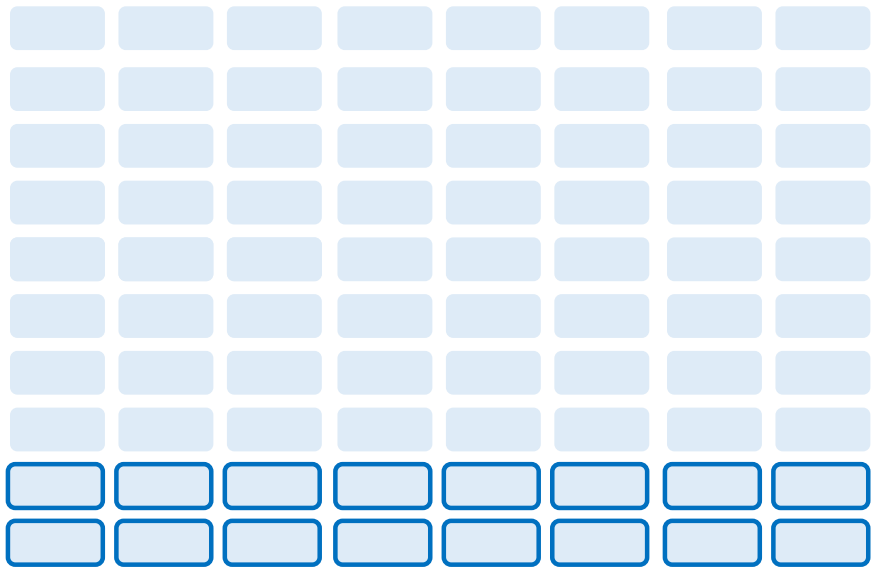

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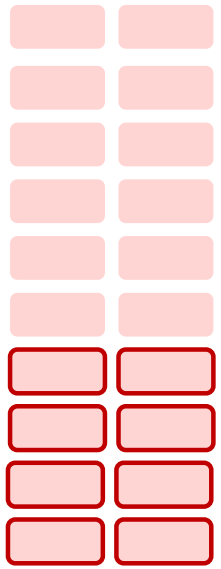
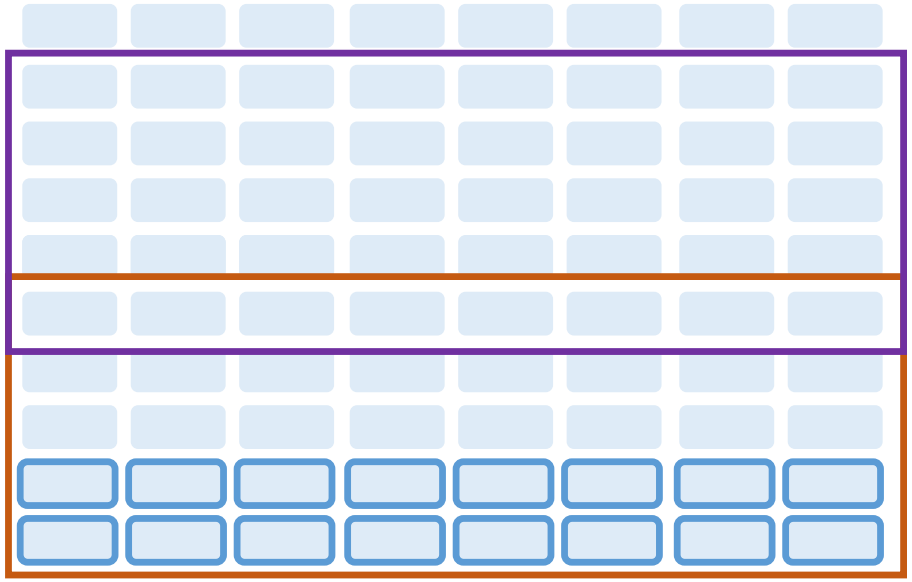


40% (8)
Red Nose

20 loan applicants,
40% will pay back

Example: Statistical or Demographic Parity

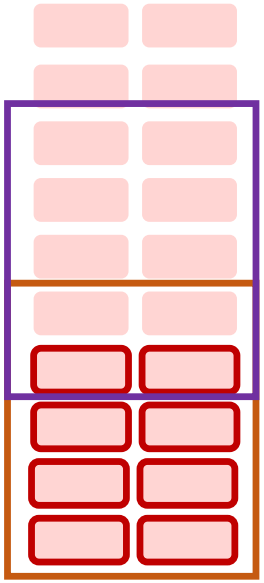
80 applicants, 20% will pay back



50% overall

Blue Nose

20 applicants, 40% will pay back



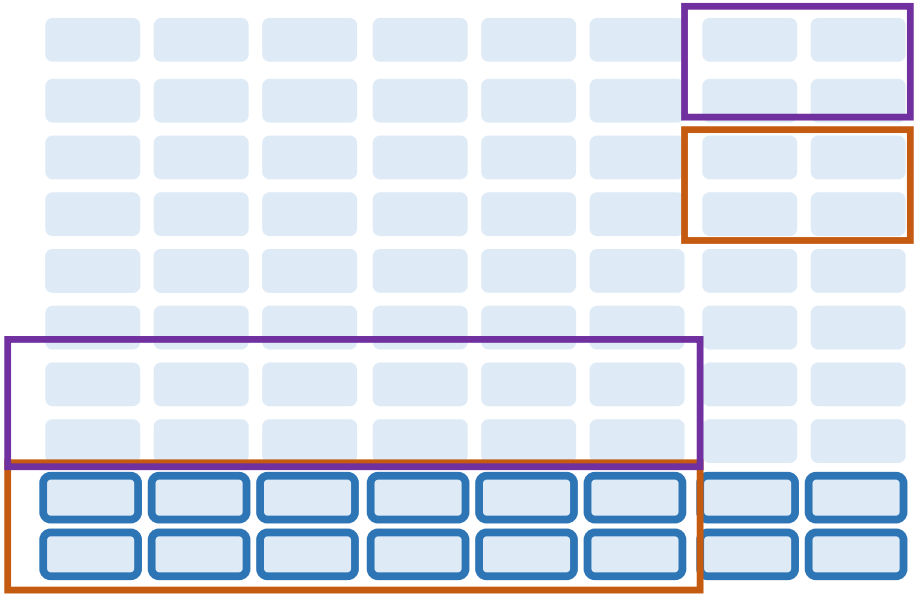
50% overall

Red Nose

Protected attribute is nose color.

Example: Equality of Positive Predictive Value

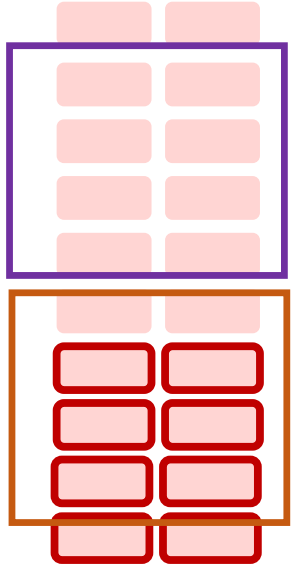
80 applicants, 20% will pay back



75% of predicted payback

Blue Nose

20 applicants, 40% will pay back



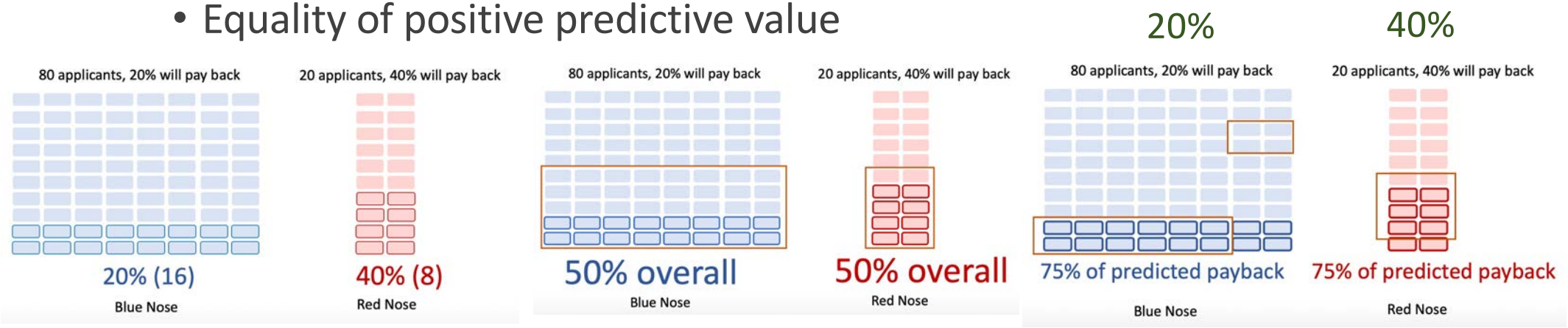
75% of predicted payback

Red Nose

Protected attribute is nose color.

Subjective Judgment: Trade-off between Fairness Definitions

- Incompatible fairness definitions:
  - Combination of Equality of both positive and negative rates across groups and
  - Equality of positive predictive value



Ref: The Ethical Algorithm, Chapter 2, 2019

Fairness

Common notions

Individual fairness: similar individuals treated similarly

Group fairness: demographic parity, equal opportunity, equalized odds

Predictive parity: equal reliability of predictions across groups

Key challenges

Different base rates and outcomes across groups

Historical and structural bias in data and labels

Proxy features can reintroduce protected attributes

Harms and benefits are context-dependent

Implications

Fairness definitions reflect different values and are often incompatible

No single metric captures fairness in all settings

In practice

Define the decision context and affected stakeholders

Choose appropriate fairness metrics and mitigation methods

Use documentation and ongoing human oversight

Key takeaway

Fairness requires explicit choices, trade-offs, and accountability—not a single optimization objective.

Refs: Russell & Norvig (2020); Hardt et al. (2016); Kleinberg et al. (2017); Barocas et al. (2023); NIST (2023)

Safety and Robustness

Examples – Toward Trusted Advice

When and what to trust?

Safety, Robustness, and Trust in AI

Key question:

When, how, and under what conditions should AI outputs be trusted?

Why this matters:

AI systems operate in dynamic, high-stakes environments

Failures may arise from uncertainty, misuse, or changing conditions

Key idea:

Trust must be earned and calibrated, not assumed.

What Is Robust and Safe AI?

Robust AI:

- Maintains acceptable performance under noise and perturbations
- Handles distribution shift and novel situations
- Degrades gracefully under partial failures

Safe AI:

- Avoids unacceptable harm, even when models are wrong
- Relies on uncertainty awareness and human oversight

Model- and data-level:

- Robust optimization and regularization
- Uncertainty estimation and risk-sensitive objectives
- Adversarial robustness within defined threat models

System-level:

- Out-of-distribution detection
- Ensemble approaches
- Hybrid or causal models where appropriate

Human-in-the-loop:

- Fail-safe mechanisms and escalation

Refs: Russell & Norvig (2020); Dietterich (2017); Bommasani et al. (2021); Weidinger et al. (2021); NIST (2023)

Validation, Monitoring, and Assurance

Validation is hard	Complex data, code, and model dependencies Feedback loops and emergent behavior Changing environments and user behavior
Good practice: ART - Accountability, responsibility, transparency	Verification: system meets specifications Validation: system meets user and safety needs Continuous monitoring post-deployment
Toward Trustworthy AI	Safety and robustness are lifecycle properties, not one-time checks.

Refs: Russell & Norvig (2020); Dietterich (2017); Bommasani et al. (2021); Weidinger et al. (2021); NIST (2023)

Governance and Regulation

Stakeholders and Responsibilities

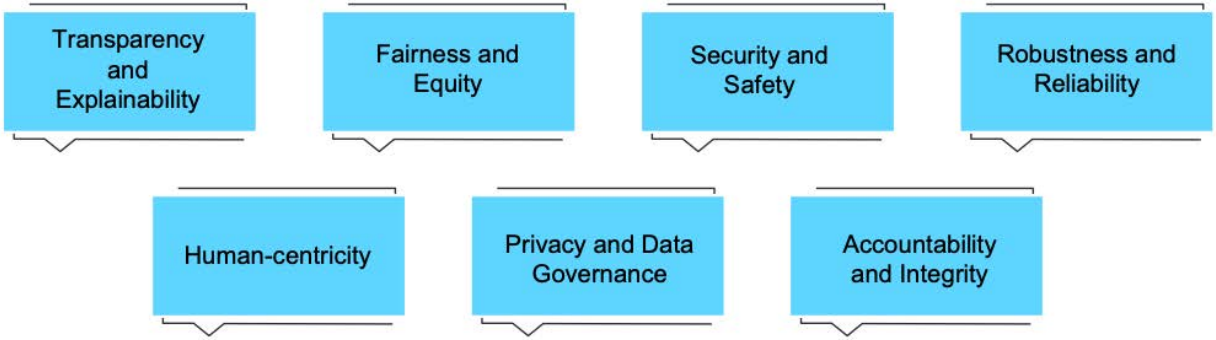
ASEAN Guide on AI Governance and Ethics (2024)

Executive Summary

What the Guide is about

This document serves as a practical guide for organisations in the region that wish to design, develop, and deploy traditional AI technologies in commercial and non-military or dual-use applications. This Guide focuses on encouraging alignment within ASEAN and fostering the interoperability of AI frameworks across jurisdictions. It also includes recommendations on national-level and regional-level initiatives that governments in the region can consider implementing to design, develop, and deploy AI systems responsibly.

Guiding Principles for the Framework



https://asean.org/book/asean-guide-on-ai-governance-and-ethics

Singapore's AI Assurance Ecosystem

- Policy & Governance Foundation
  - Model AI Governance Framework (2nd Edition) – Personal Data Protection Commission (PDPC)
  - Principles: Explainable, transparent, fair, and human-centric AI
  - Practice areas: Internal governance, human oversight, operations management, stakeholder communication
    - Framework (PDF): https://www.pdpc.gov.sg/-/media/files/pdpc/pdf-files/resource-for-organisation/ai/sgmodelaigovframework2.pdf
    - Overview: https://www.pdpc.gov.sg/help-and-resources/2020/01/model-ai-governance-framework
- National Strategy
  - National AI Strategy 2.0 (NAIS 2.0)
  - Whole-of-nation AI roadmap (launched Dec 2023)
  - Priorities: Talent, compute, industry adoption, safety-by-design
  - Goal: Build a trusted, cross-sector AI ecosystem
    - Strategy: https://www.smartnation.gov.sg/initiatives/national-ai-strategy/
- Assurance & Safety Infrastructure
  - **AI Verify Foundation**
  - Open-source AI testing and assurance toolkit
  - Supports global standards (e.g. ISO/IEC 42001)
  - Public-private partnership for scalable AI assurance
    - Official link: https://aiverifyfoundation.sg
  - **Singapore AI Safety Institute (AISI)**
  - National institute for AI evaluation science and red-teaming
  - Focus on frontier model safety and international collaboration
    - Official link: : https://www.sgaisi.sg

AI Regulation on the Horizon

https://www.gov.cn/zhengce/202307/content_6892001.htm

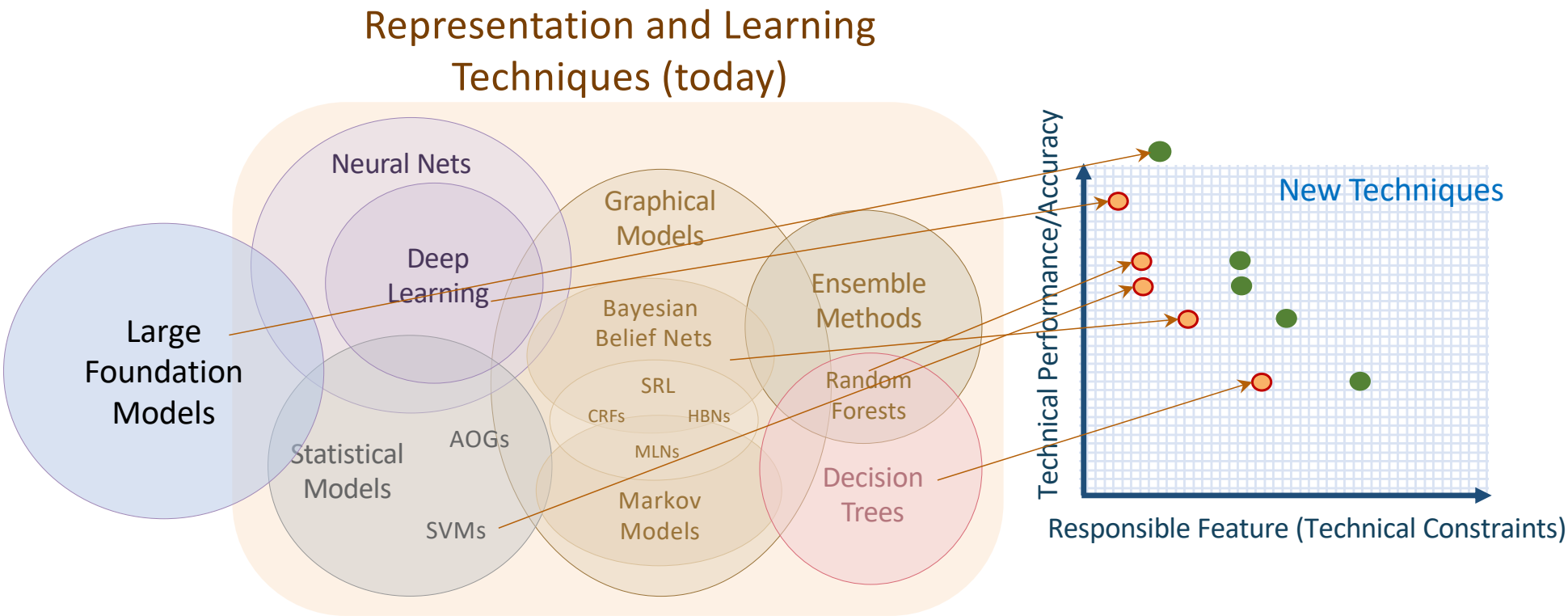
https://www.whitehouse.gov/ostp/ai-bill-of-rights/

Source:
https://www.hsa.gov.sg/docs/default-source/hprg-mdb/gudiance-documents-for-medical-devices/regulatory-guidelines-for-software-medical-devices---a-life-cycle-approach.pdf

https://www.europarl.europa.eu/news/en/headlines/society/20230601STO93804/eu-ai-act-first-regulation-on-artificial-intelligence

https://www.nature.com/articles/d41586-023-02491-y

Performance vs Responsible Feature



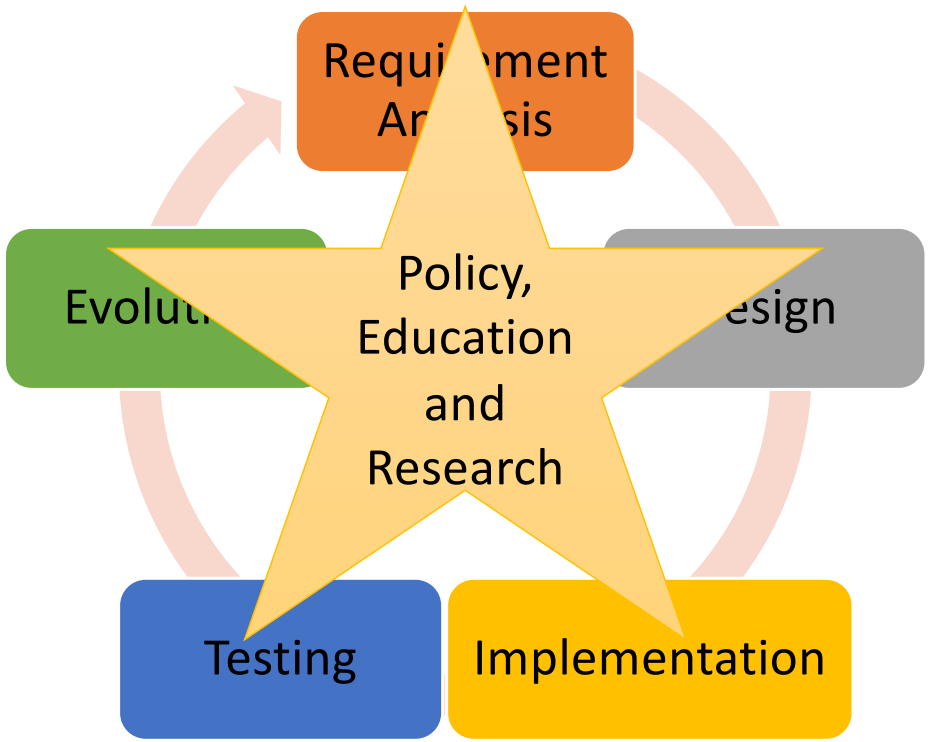
Adapted from: Resources on https://www.darpa.mil/program/explainable-artificial-intelligence

Making Responsible Decisions

- Core issue
  - Balance accuracy with responsible features -- transparency, privacy, fairness, safety and robustness, human factors, and governance.
- Examples:
  - Diagnosis recommendations, risk predictions, optimal policies, assistive guidance, recruitment interviews, social assistance schemes, university admissions, population censors , consumer behaviors, scientific research
- Key questions
  - What responsible feature matters here?
  - Why is it important in this context?
  - How can trust in the AI system be ensured?
  - What tools and techniques are available?
  - What are the trade-offs with accuracy?
  - What are the implications for stakeholders? Developer, user, or owner/manager/regulator
  - Who decides, and when in the system lifecycle?

When to Make Responsible Decisions?

- In AI design and application?
- In software engineering process?
- In deployment and maintenance?
- Who should make decisions?
- What are current practices?



Summary

- Toward Responsible AI
  - Balance accuracy with other considerations.
  - Consider human, ethical, governance, and regulatory factors.
  - Some scientific methods are ready for use; others are still evolving.
  - Human judgment must guide AI throughout its lifecycle.
- Take-home ideas
  - Use human-aware approaches to develop and deploy AI.
  - Combine human policies with scientific methods.

Homework

- Watch the Videos:
  - **Thinking Ethics in AI Series**| UNESCO[1]
    - (4 short videos – first 2 videos are different versions of the same video)
    - Youtube: https://youtube.com/playlist?list=PLWuYED1WVJINThtWMB4QbVNQ1yiTxxGf6
  - **How artificial intelligence is changing our society** | DW Documentary [Apr 2020]
    - Youtube: https://youtu.be/-ePZ7OdY-Dw
- Experiment with ChatGPT or LLAMA or DeepSeek or QWEN:
  - https://openai.com/blog/chatgpt/
    https://ai.meta.com/llama/
  - https://www.deepseek.com
    https://chat.qwen.ai
- Readings:
  - RN: 27.1, 27.3; *27.2 (optional)*
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